Vector addition

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add displacement vectors and use a diagram or equation to explain where an object ends up.

SUCCESS CRITERIA - I can...

1. Draw and label arrows tip to tail; keep their length and direction.
2. Draw the resultant from the original start to the final endpoint.
3. State net east-west and north-south motion with units and direction.

Core: teacher selects from 1-10. **Stretch:** 11-12 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “vector” and “component” to explain your answer. Sentence starters are optional.

vector: a quantity with size and direction

component: the part of a vector along one chosen axis

resultant: one vector equal to the sum of other vectors

01 Warm-up: bring back an earlier idea

A cart moves from **0 m** to **8 m** *east* in **4 s** at a steady pace.
1. **Find its speed**.
2. **Draw a velocity arrow showing** its direction.
3. **Explain what the arrow tells** you that speed alone does not.

Word help: velocity = rate of position change, with direction

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Read and explain a worked example

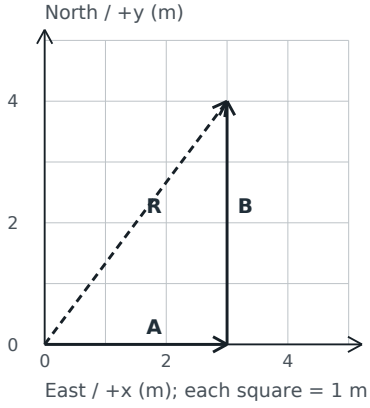
Reference: $R_x = A_x + B_x$; $R_y = A_y + B_y$.
Use *east* = +x and *north* = +y. **Add signed components**; **draw R** from the original start to the final endpoint.

Worked reasoning: Add *east*-*west* and *north*-*south* motion separately: R_x = **3 m** and R_y = **4 m**. The route length is **3** + **4** = **7 m**. The supplied straight-line length is **5 m**.

Your turn: A walker goes **3 m** *east*, then **4 m** *north*. On the grid, circle the original start and final endpoint. **Explain why the distance** is **7 m** while the displacement magnitude is **5 m**.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I think … . I can point to …, which shows …, so … .



Marzano check - circle: ① ② ③

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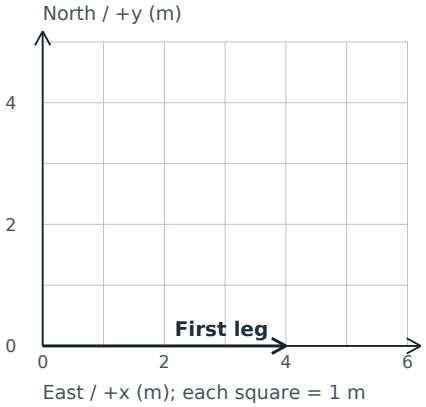
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Finish the diagram or calculation

Walk **4 m** *east*, then **3 m** *north*. The first arrow is drawn. **Add the second arrow** from its tip and a dashed resultant from the original start.

Sentence starter: In my work, I used … to show … . I labeled it … .

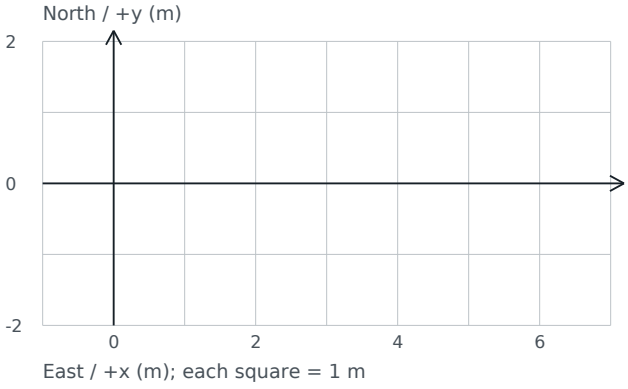


Marzano check - circle: ① ② ③

04 A trip that doubles back

Walk **6 m** *east*, then **4 m** *west*.
1. Draw and **label the route** and resultant.
2. **Give distance** and displacement.
3. **Explain which length follows the** path.

Sentence starter: I think … . I can point to …, which shows …, so … .

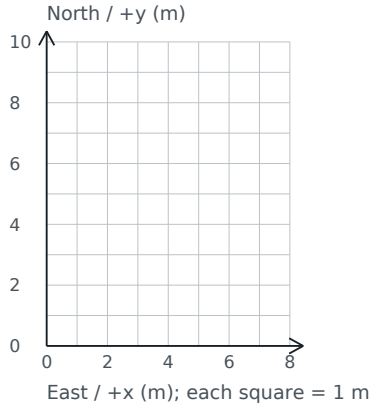


Marzano check - circle: ① ② ③

05 Use a new scale

Walk **6 m** *east*, then **8 m** *north*. **Draw both arrows** and the resultant. **Give distance** and displacement. The straight-line length is supplied: **10 m**.

Sentence starter: In my work, I used … to show … . I labeled it … .

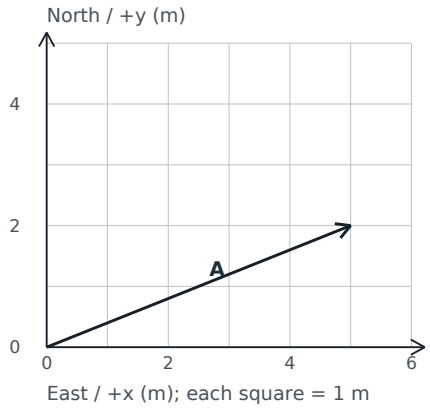


Marzano check - circle: ① ② ③

06 Add two diagonal arrows

A: **5 m** *east* and **2 m** *north*. B: **3 m** *west* and **1 m** *north*. A is drawn.
1. **Add B** and R.
2. **State each net component** and the general compass direction.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

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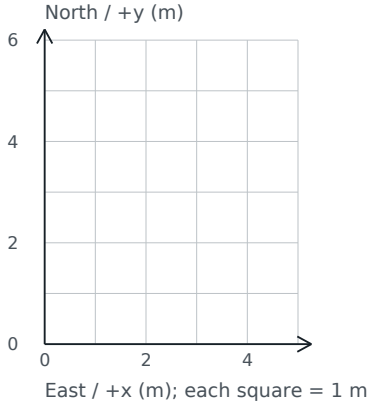
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Predict, draw, check

Walk **3 m** *east*, **4 m** *north*, then **3 m** *west*.
1. **Predict what cancels**.
2. **Draw the route** and R.
3. **Give distance** and displacement.
4. **Check your prediction against the** grid.

Sentence starter: I predict … . I used … to make this prediction because … .

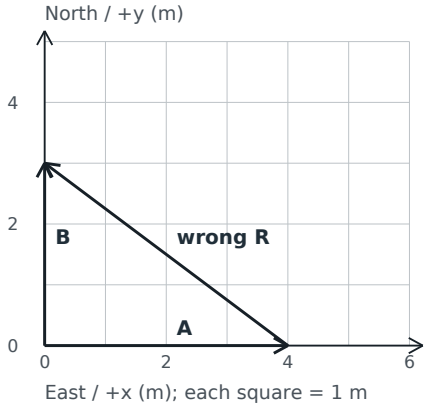


Marzano check - circle: 1 2 3

08 Find and repair the mistake

Sam draws the two displacement arrows from the same start, then connects their tips and calls that arrow the resultant. The grid shows his mistake.
1. **Draw the correct tip-to-tail route** beside it.
2. **Explain where the correct resultant** must begin and end.

Sentence starter: The step I would change is … . I would change it because I found … .



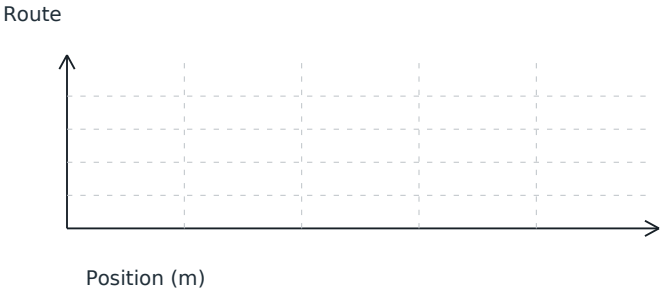
Marzano check - circle: 1 2 3

09 One trip, three questions

A student walks **6 m** *east* in **6 s**, then **8 m** *north* in **8 s**. The direct start-to-finish length is **10 m**. For each answer, decide whether to use path length or the start-to-finish arrow, then solve:
(a) Distance walked? (b) Displacement?
(c) Average speed for the whole trip?
Label the diagram with the given positions and any prediction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I predict … . I used … to make this prediction because … .



Marzano check - circle: 1 2 3

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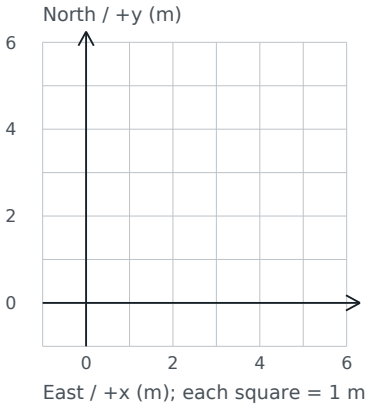
Continue down each column.

Sentence starters are optional. You may plan in any language.

10 Exit check - work independently

A robot makes two moves. First: **4 m** *east* and **1 m** *north*. Second: **1 m** *west* and **3 m** *north*.
1. **Draw both moves tip-to-tail** and **add the resultant**.
2. **State its two components** and general compass direction.
3. **Explain one way** to **check the result**.

Sentence starter: I think … . I can point to …, which shows …, so … .

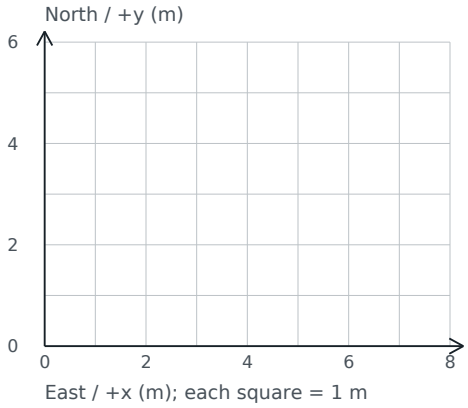


Marzano check - circle: ① ② ③

11 STRETCH: Design two routes

Finish **4 m** *east* and **3 m** *north* of your start.
1. Design one route of length **7 m** and another of length **11 m**.
2. Use only *north*, *south*, *east* or *west* legs.
3. **Label all lengths** and **draw each resultant**.

Sentence starter: My design includes … . It does what is needed by … .



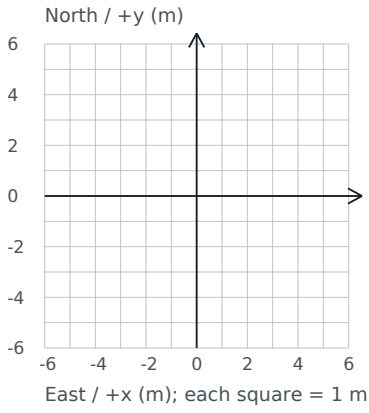
Marzano check - circle: ① ② ③

12 STRETCH: Explore the limits

1. Point two **5 m** displacement arrows in any directions.
2. **Draw the smallest** and largest possible resultants, then one between those extremes.
3. **Estimate the intermediate resultant** and **explain why your limits make** sense.

Word help: estimate = find an approximate value

Sentence starter: I think … . I can point to …, which shows …, so … .



Marzano check - circle: ① ② ③